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WPC POWER DRIVER BOARD – COST–OPTIMISED MODERN VERSION – functional replacement for Williams A–12697–3 (same outline, connectors, fuses).
DPAK MOSFET solenoid/lamp drivers, TPS54360B buck regulators for +5V/+12V, 4–quadrant insulated triacs, SMD passives, 5x20 mm fuses.
Sheets: Power_Supply, CPU_Interface, Sol_HighPower (1–8), Sol_LowPower (9–16), Sol_Flashers (17–24), Sol_GeneralPurpose (25–28),
Lamp_Columns, Lamp_Rows, GI_Triacs. Every part carries Manufacturer / MPN / DigiKey link / Price / Price100 fields (see BOM). See README.md.

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CPU_Interface

File: cpu_interface.kicad_sch

Sol_HighPower

File: sol_highpower.kicad_sch

Sol_LowPower

File: sol_lowpower.kicad_sch

Sol_Flashers

File: sol_flashers.kicad_sch

Sol_GeneralPurpose

File: sol_gp.kicad_sch

Lamp_Columns

File: lamp_columns.kicad_sch

Lamp_Rows

File: lamp_rows.kicad_sch

GI_Triacs

File: gi_triacs.kicad_sch

Modern re–implementation of the Williams A–12697–3 (same form factor)

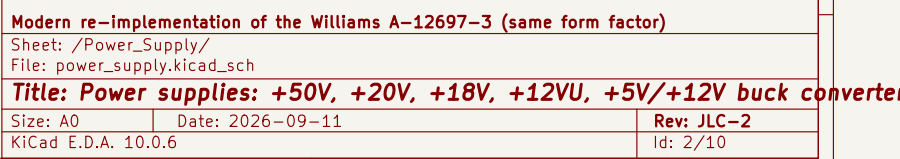
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Title: WPC Power Driver Board – cost–optimised modern re–implementation

Size: A3Date: 2026–09–11Rev: JLC–2

KiCad E.D.A. 10.0.6Id: 1/10

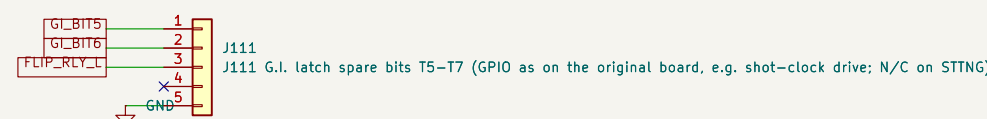
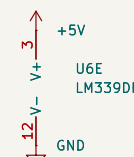
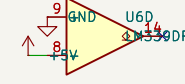
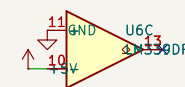
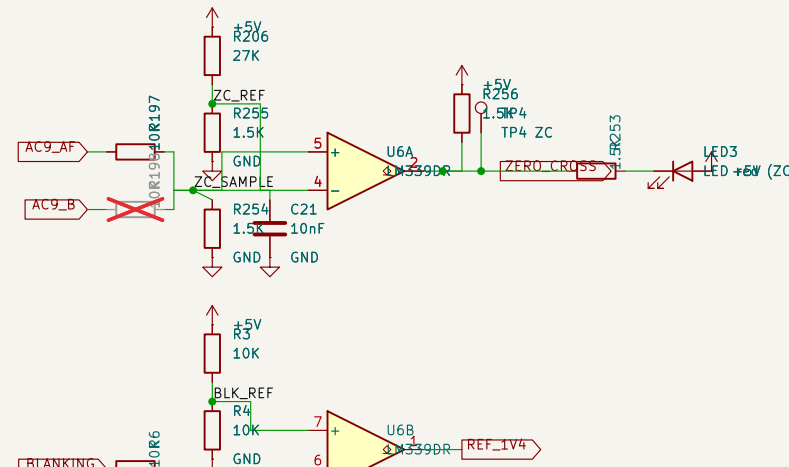
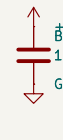
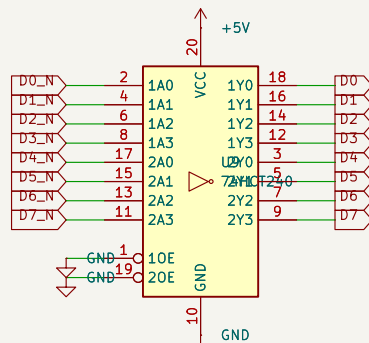
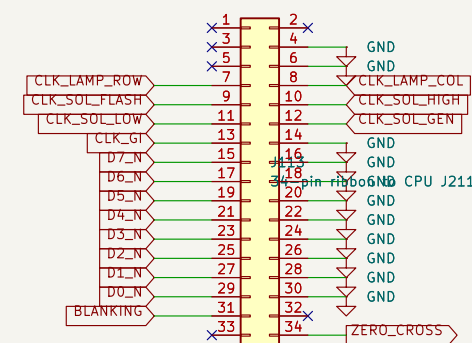
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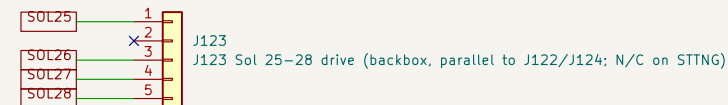
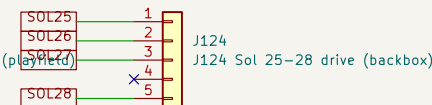
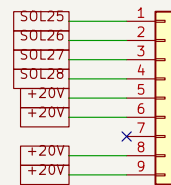
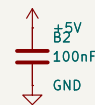
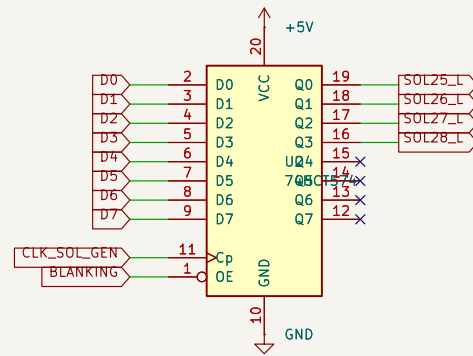
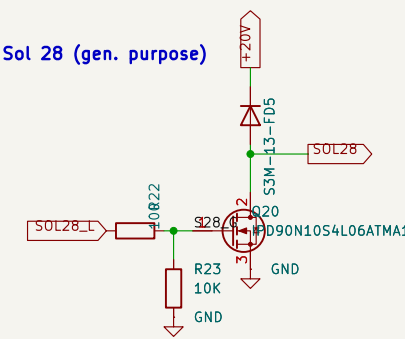
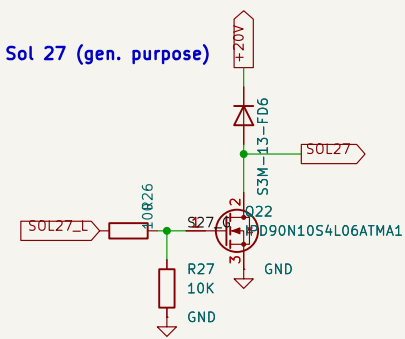
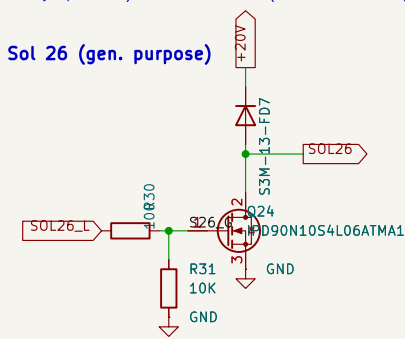
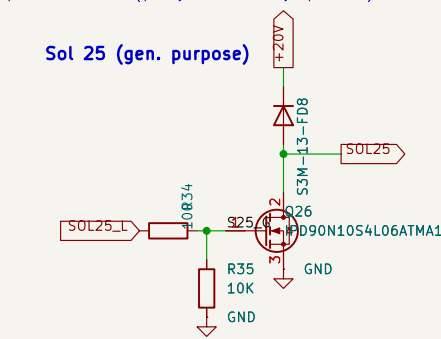
CPU interface, pin-out from the WPC schematic manual 16-9834.2. J113: 7 /LMP ROW, 8 /LMP COL, 9 /SOL 2 (sol 17-24), 10 /SOL 4 (sol 1-8), 11 /SOL 3 (sol 9-16)
12 /SOL 1 (sol 25-28), 13 /TRIAC, 15..22 odd = /D7../D0, 31 BLANKING (HIGH = every latch tri-stated, open-collector from the CPU, pulled up here),
34 ZENER CROSS to the CPU, 1 /V+ 5V, 2..27 odd = address output of the 4 yellow boards, and 3 even ALGCS, then A 12603, 31 ethereal pins unassigned.

34 ZERO CROSS to the CPU; 1/3/5/32/33 (flipper-opto sense of the -1 relay board) and 2 are N/C on the A-12697-3; other even pins = ground.
The CPU board drives the data through a 74LS240, so the ribbon data is ACTIVE LOW: U9 (74HCT240) inverts it and every latch then takes the active low data. The CPU also has a 74LS240 (U8) which provides the blanking signal. Blankings are taken from 74HCT571 (make good connections to the CPU board).
The CPU board also has a 74LS240 (U7) which provides the blanking signal. Blankings are taken from 74HCT571 (make good connections to the CPU board).

The true bus (register bit 1 → output ON, assign NAME / FreePac write it). BLANKING tri-states every 74HC1574 (gate 9) at 0.6 V reference level. The lamp—row 1, 4 V reference low. Zero CROSS: the F113—fused leg of the 9 VAC winding is sampled through a resistor network (R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R74, R75, R76, R77, R78, R79, R80, R81, R82, R83, R84, R85, R86, R87, R88, R89, R90, R91, R92, R93, R94, R95, R96, R97, R98, R99, R100, R101, R102, R103, R104, R105, R106, R107, R108, R109, R110, R111, R112, R113, R114, R115, R116, R117, R118, R119, R120, R121, R122, R123, R124, R125, R126, R127, R128, R129, R130, R131, R132, R133, R134, R135, R136, R137, R138, R139, R140, R141, R142, R143, R144, R145, R146, R147, R148, R149, R150, R151, R152, R153, R154, R155, R156, R157, R158, R159, R160, R161, R162, R163, R164, R165, R166, R167, R168, R169, R170, R171, R172, R173, R174, R175, R176, R177, R178, R179, R180, R181, R182, R183, R184, R185, R186, R187, R188, R189, R190, R191, R192, R193, R194, R195, R196, R197, R198, R199, R200, R201, R202, R203, R204, R205, R206, R207, R208, R209, R210, R211, R212, R213, R214, R215, R216, R217, R218, R219, R220, R221, R222, R223, R224, R225, R226, R227, R228, R229, R230, R231, R232, R233, R234, R235, R236, R237, R238, R239, R240, R241, R242, R243, R244, R245, R246, R247, R248, R249, R250, R251, R252, R253, R254, R255, R256, R257, R258, R259, R260, R261, R262, R263, R264, R265, R266, R267, R268, R269, R270, R271, R272, R273, R274, R275, R276, R277, R278, R279, R280, R281, R282, R283, R284, R285, R286, R287, R288, R289, R290, R291, R292, R293, R294, R295, R296, R297, R298, R299, R300, R301, R302, R303, R304, R305, R306, R307, R308, R309, R310, R311, R312, R313, R314, R315, R316, R317, R318, R319, R320, R321, R322, R323, R324, R325, R326, R327, R328, R329, R330, R331, R332, R333, R334, R335, R336, R337, R338, R339, R340, R341, R342, R343, R344, R345, R346, R347, R348, R349, R350, R351, R352, R353, R354, R355, R356, R357, R358, R359, R360, R361, R362, R363, R364, R365, R366, R367, R368, R369, R370, R371, R372, R373, R374, R375, R376, R377, R378, R379, R380, R381, R382, R383, R384, R385, R386, R387, R388, R389, R390, R391, R392, R393, R394, R395, R396, R397, R398, R399, R400, R401, R402, R403, R404, R405, R406, R407, R408, R409, R410, R411, R412, R413, R414, R415, R416, R417, R418, R419, R420, R421, R422, R423, R424, R425, R426, R427, R428, R429, R430, R431, R432, R433, R434, R435, R436, R437, R438, R439, R440, R441, R442, R443, R444, R445, R446, R447, R448, R449, R450, R451, R452, R453, R454, R455, R456, R457, R458, R459, R460, R461, R462, R463, R464, R465, R466, R467, R468, R469, R470, R471, R472, R473, R474, R475, R476, R477, R478, R479, R480, R481, R482, R483, R484, R485, R486, R487, R488, R489, R490, R491, R492, R493, R494, R495, R496, R497, R498, R499, R500, R501, R502, R503, R504, R505, R506, R507, R508, R509, R510, R511, R512, R513, R514, R515, R516, R517, R518, R519, R520, R521, R522, R523, R524, R525, R526, R527, R528, R529, R530, R531, R532, R533, R534, R535, R536, R537, R538, R539, R540, R541, R542, R543, R544, R545, R546, R547, R548, R549, R550, R551, R552, R553, R554, R555, R556, R557, R558, R559, R560, R561, R562, R563, R564, R565, R566, R567, R568, R569, R570, R571, R572, R573, R574, R575, R576, R577, R578, R579, R580, R581, R582, R583, R584, R585, R586, R587, R588, R589, R590, R591, R592, R593, R594, R595, R596, R597, R598, R599, R600, R601, R602, R603, R604, R605, R606, R607, R608, R609, R610, R611, R612, R613, R614, R615, R616, R617, R618, R619, R620, R621, R622, R623, R624, R625, R626, R627, R628, R629, R630, R631, R632, R633, R634, R635, R636, R637, R638, R639, R640, R641, R642, R643, R644, R645, R646, R647, R648, R649, R650, R651, R652, R653, R654, R655, R656, R657, R658, R659, R660, R661, R662, R663, R664, R665, R666, R667, R668, R669, R670, R671, R672, R673, R674, R675, R676, R677, R678, R679, R680, R681, R682, R683, R684, R685, R686, R687, R688, R689, R690, R691, R692, R693, R694, R695, R696, R697, R698, R699, R700, R701, R702, R703, R704, R705, R706, R707, R708, R709, R710, R711, R712, R713, R714, R715, R716, R717, R718, R719, R720, R721, R722, R723, R724, R725, R726, R727, R728, R729, R730, R731, R732, R733, R734, R735, R736, R737, R738, R739, R740, R741, R742, R743, R744, R745, R746, R747, R748, R749, R750, R751, R752, R753, R754, R755, R756, R757, R758, R759, R760, R761, R762, R763, R764, R765, R766, R767, R768, R769, R770, R771, R772, R773, R774, R775, R776, R777, R778, R779, R780, R781, R782, R783, R784, R785, R786, R787, R788, R789, R790, R791, R792, R793, R794, R795, R796, R797, R798, R799, R800, R801, R802, R803, R804, R805, R806, R807, R808, R809, R810, R811, R812, R813, R814, R815, R816, R817, R818, R819, R820, R821, R822, R823, R824, R825, R826, R827, R828, R829, R830, R831, R832, R833, R834, R835, R83

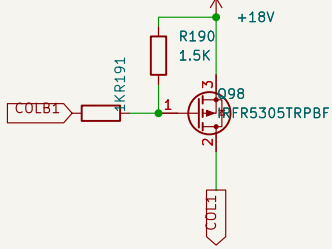


General purpose drivers (sol. 25–28): IPD90N10S4L–06 DPAK, latch U2 output HIGH= ON, clocked by /SOL 1 (J113–12). S3M tie-back diode cathodes are brought out to J122 pins 5,6,8,9 so the harness can tie them to the supply actually used (+50V coil or +20V flasher). Outputs J122 (playfield, key pin 7), J123 (backbox, key pin 2) and J124 (cabinet, key pin 4) as on the A–12697. U2 bits 4–7 unused.

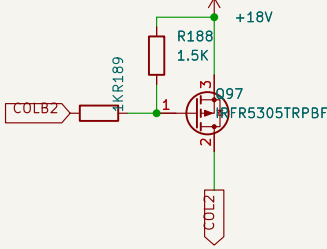


Lamp matrix column drivers. Column strobe register 0x3FE5: bit(k-1)=1 selects column k, clocked by /LMP COL (J113-8). The TBD62083A inputs have an internal pull-down (Toshiba usage note), so a tri-stated U18 (BLANKING) leaves every column OFF - the OEM 74LS374/ULN2803 pair behaves the same way. U18 latches the data bus; the TBD62083A DMOS sink pulls the P-MOSFET gate down through 1 k against the 1.5 k gate-source resistor (Vgs = -0.6 x V+18 = -8..-12 V) -> column high (+18V) with -65 mOhm (IRFR5305 DPAK; TIP107 dropped -1.2 V). Column k = Q(99-k): col1=Q98 ... col8=Q91. Outputs J137/J138 pins 1-7,9 (key pin 8).

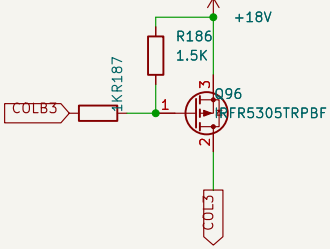
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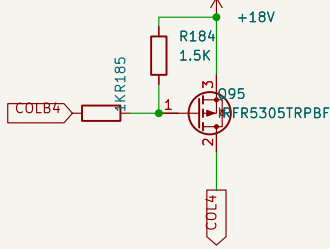
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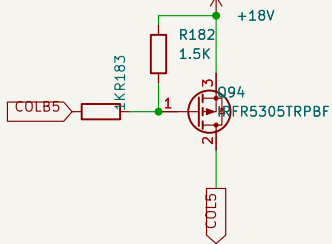
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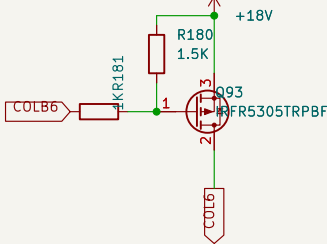
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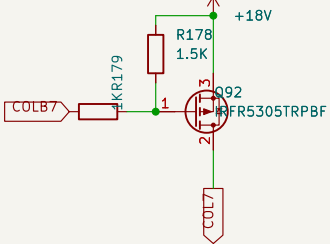
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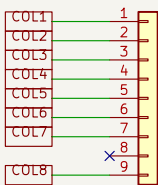
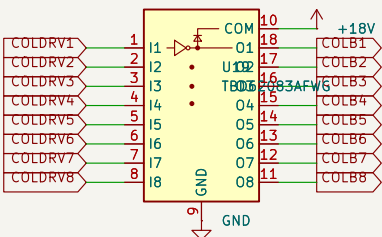
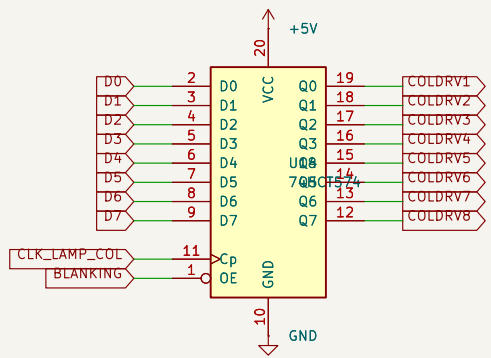
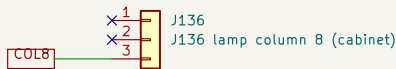
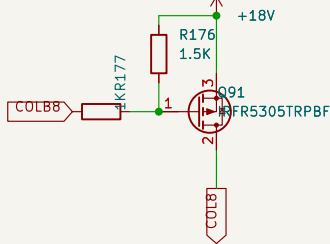
Lamp column 6



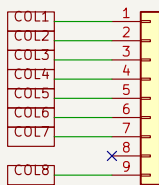
Lamp column 7



Lamp column 8



J137 lamp columns (playfield)



J138 lamp columns (backbox)

Modern re-implementation of the Williams A-12697-3 (same form factor)

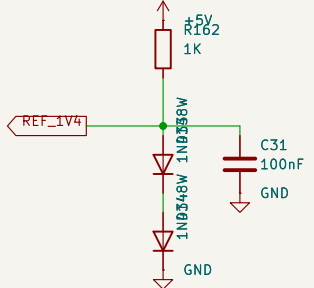
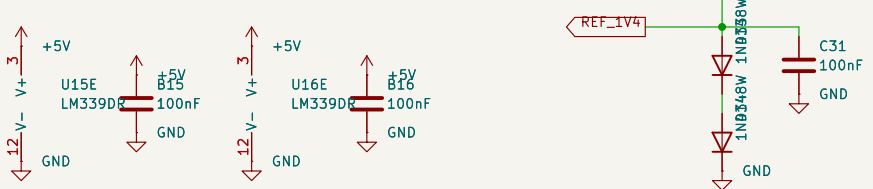
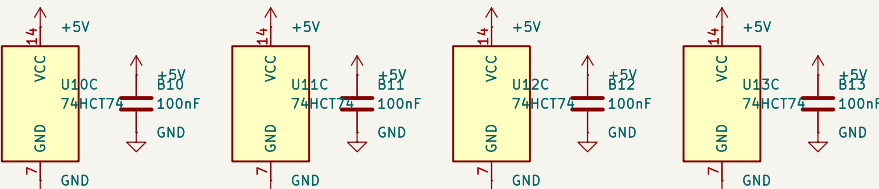
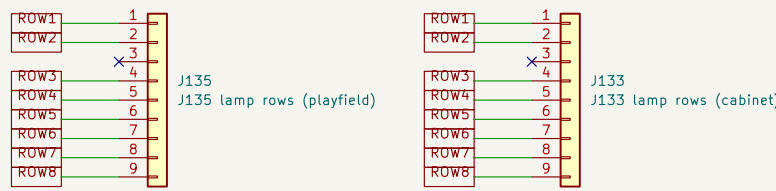
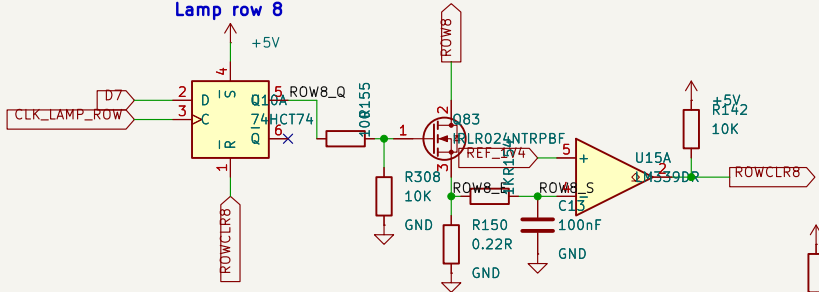
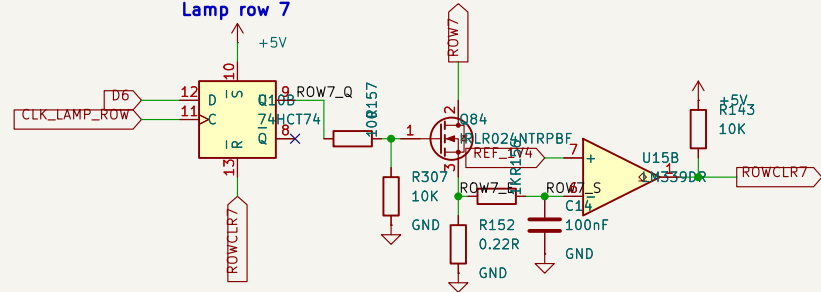
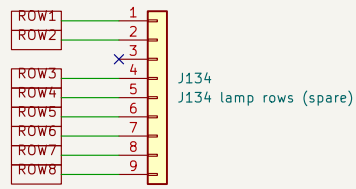
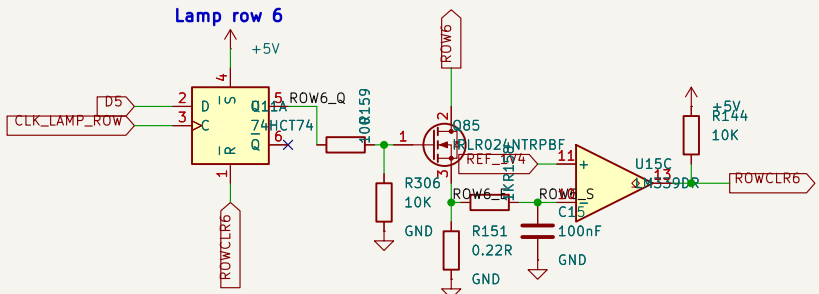
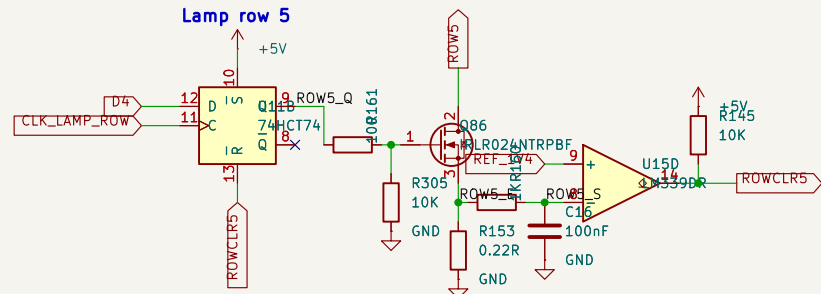
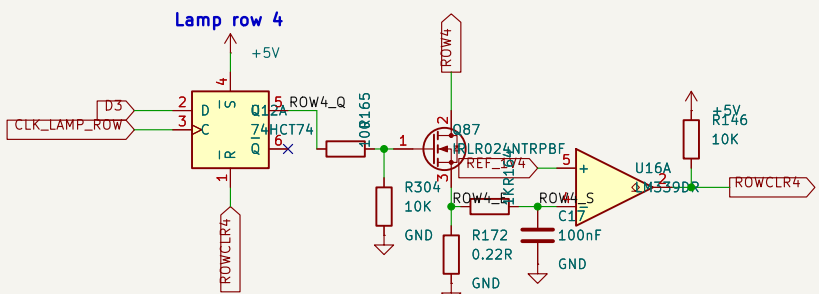
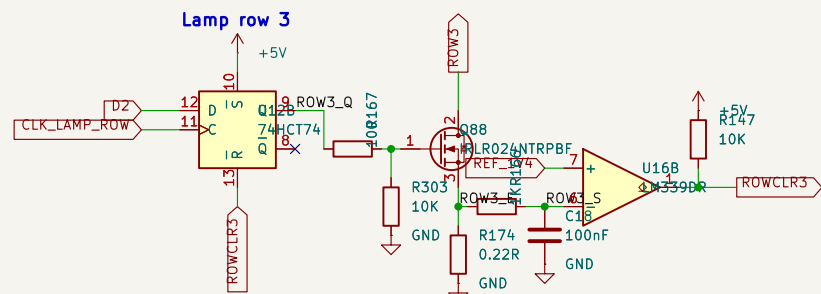
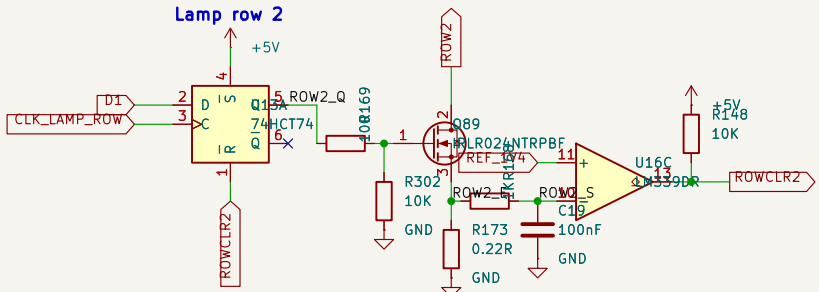
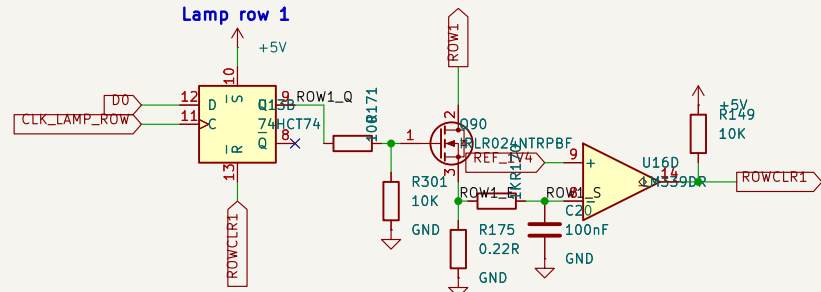
Sheet: /Lamp_Columns/
File: lamp_columns.kicad_sch

Title: Lamp matrix columns (IRFR5305 high side, +18V)

Size: A2 Date: 2026-09-11
KiCad E.D.A. 10.0.6

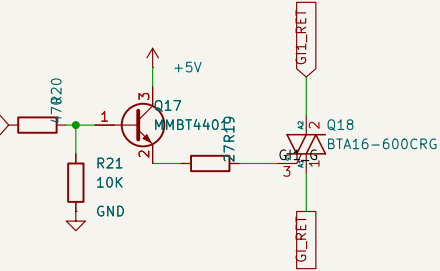
Rev: JLC-2
Id: 8/10

Lamp matrix row drivers. Row register 0x3FE4: bit(k-1)=1 turns row k on, clocked by /LMP ROW (J113-7). The 74HCT74 latches the data bit (true after U9): Q high -> IRLR024N on (row = ground, -65-80 mOhm DPAK). Over-current: source sense 0.22 R -> LM339; above the 1.4 V reference (6 A) the open-collector output pulls -CLR low and the row is shut off until the next strobe. BLANKING pulls REF_1V4 low (CPU sheet) so all rows are forced off. Row k = Q(91-k): row1=Q90 ... row8=Q83. Outputs J133/J135 (pins 1,2,4-9; key 3).

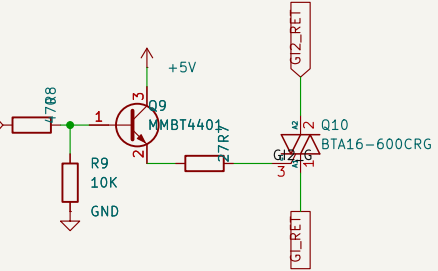


General illumination: each 6.3VAC string is fused (F106–F110) on the hot side and switched by a triac in the RETURN path; MT1 of the five triacs form the G_LRET bus that goes back to the transformer on J115–7/8/10/11/12 and is referenced to logic ground by W1 (OEM: J115–1 GND REF), so the 5 x 5 A return current stays off the ground plane. Latch U1 (clocked by TRIAC, J113–13) output HIGH = string ON. The gate is fed by an NPN emitter follower from +5V through 27 R (–80 mA, positive gate current in quadrants I and IV → 4-quadrant “C”-grade triac; snubberless BW/CW types cannot fire in Q IV). 10 k base pull-downs keep the strings OFF while the latch is tri-stated by BLANKING. Bits 5–7 (T5–T7) go to J111 (GPIO, CPU sheet); bit 7 also drives the optional flipper relay (DNP).

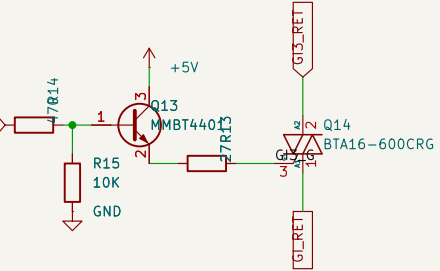
G.I. string #1 (Wht–Brn / Brn)



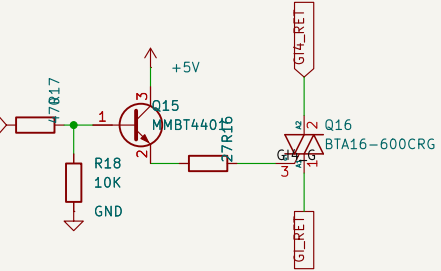
G.I. string #2 (Wht–Org / Org)



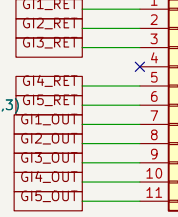
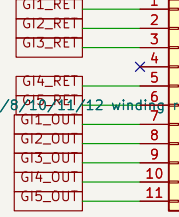
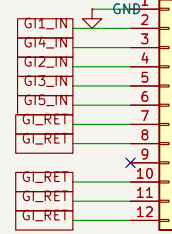
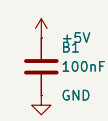
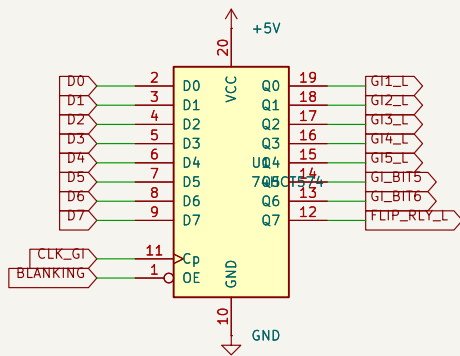
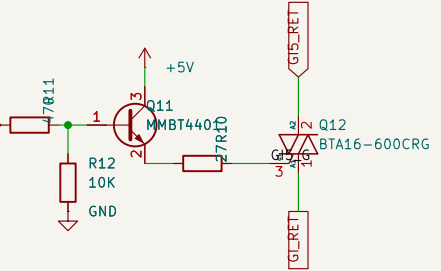
G.I. string #3 (Wht–Yel / Yel)



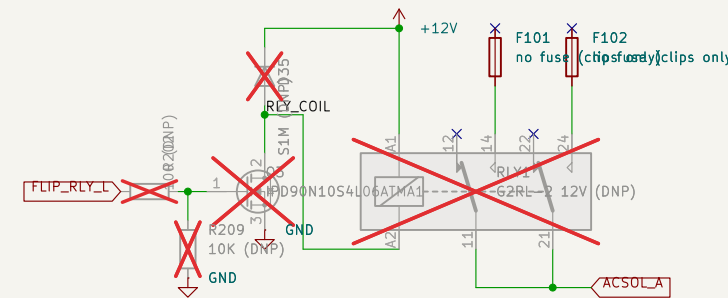
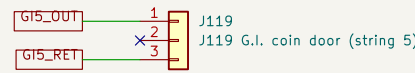
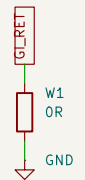
G.I. string #4 (Wht–Grn / Grn)



G.I. string #5 (Wht–Vio / Vio)



J121 J122 G.I. strings 1–5 (playfield harness uses strings 1,4,5)



Modern re-implementation of the Williams A-12697-3 (same form factor)

Sheet: /G.I./Triacs/
File: g.l.triacs.kicad.sch

Title: General illumination triacs + flipper relay option

Size: A1 Date: 2026-09-11
Kicad E.D.A. 10.0.6

Rev: JLC-2
10/10/25